

# The empirical recurrence for OEIS A210058, proved by counting patterns

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## Abstract

OEIS A210058 counts arrays subject to a local condition *and* to the clause “new values 0..3 introduced in row major order”, which says the array is written in canonical form. That clause is not local, so a transfer matrix over the alphabet does not see it. It does not have to: what is being counted is not arrays but equality patterns, and the number of patterns is recovered from the numbers  $L_i$  of arrays over an  $i$ -letter alphabet by inverting a falling factorial. The result is the clean identity  $4!a(n) = \sum_i \binom{4}{i} D_{4-i} L_i(n)$  with  $D_m$  the derangement numbers, and each  $L_i$  is an ordinary walk count. A second reduction — the chain is strongly lumpable over the patterns, because the condition is relabelling-invariant — brings the state space down from 4890 to  $S = 342$ . The entry’s empirical recurrence of order 82, contributed by R. H. Hardin, then follows from a finite exact computation.

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## 1 The sequence and the conjecture

OEIS A210058 is “Number of  $(n+1) \times 6$  0..3 arrays with every  $2 \times 2$  subblock having one, two or four distinct values, and new values 0..3 introduced in row major order.”. It has offset 1 and begins

15435, 1800963, 216648913, 26586250017, 3291534980629, 409244238764475, 50984849090449545, 6357981795575

The entry carries the following comment:

Empirical:  $a(n) = 475*a(n-1) - 97955*a(n-2) + 11644948*a(n-3) - 886260089*a(n-4) + 44834024531*a(n-5) - 1480016997447*a(n-6) + 27501893634440*a(n-7) - 19154281755615*a(n-8) - 14109997538202492*a(n-9) + 380139811212670895*a(n-10) - 3633142641071106396*a(n-11) - 39963611616830248751*a(n-12) + 1593444365962381088141*a(n-13) - 16360359898646925132279*a(n-14) - 53628797960005666834153*a(n-15) + 3065608736301301696033933*a(n-16) - 26133079154765866041703954*a(n-17) - 87110150856708548490003060*a(n-18) + 3285447151642843946296465482*a(n-19) - 18500796538252035362244196167*a(n-20) - 113537822015480997084820017660*a(n-21) + 1969595973555246112226429400634*a(n-22) - 5094180777791326840894878530663*a(n-23) - 75410170517326615792515199294977*a(n-24) + 619290284616722614925*a(n-25) + 282521833961316282723684931217991*a(n-26) - 24485953057692811284051179308463142*a(n-27) + 87528837132342708300013321879388542*a(n-28) + 421839879798739258065978672006583145*a(n-29) - 3822531315339042347338715171300344808*a(n-30) + 1642585670325595563732651321348901955*a(n-31) + 80095320020976756490928448282418797830*a(n-32) - 249661247879983704468179465630174148693*a(n-33) - 824995758310594778361056511706446456928*a(n-34) + 6016785610935426654223191102098135317833*a(n-35) - 778697358933658882992721546717012979601*a(n-36) - 79309986916314128726892179039528969987707*a(n-37) + 157555334900999055527104579177409467289432*a(n-38) + 602283879827287863757856564608666530725055*a(n-39) - 2546742900036971451117990401862557640143168*a(n-40) - 1657027255259397066080778818344036441460639*a(n-41) + 23255596150507593709470173141631460142244853*a(n-42) - 175945423122062908435349998665066887176802*a(n-43) - 136255374929980512719104232302331973426434040*a(n-44) + 25872557454814549489220002555351976933063$

45) +491079674149355651944535333752397059873317488\*a(n-46) -17665756723540764008133525484962690515475  
 47) -683150248115121120383647344175928104099477608\*a(n-48) +78163718760003679473950388963503169679412  
 49) -3302546155481051758244037513755926584253348544\*a(n-50) -23915833694228189943256241814306852903473  
 51) +2508864390546642571395488207191547213688830368\*a(n-52) +50427907621483887112073549360396299946  
 53) -88008584836161707485167757433011709683981331136\*a(n-54) -6775442280470026605439025054695954212005  
 55) +202406445124041566579135038714540269024683517568\*a(n-56) +3748586239356461284117485526710596859  
 57) -329415495615585324293996620119764707645766443008\*a(n-58) +55853541413444206557214362404731022974  
 59) +388475434686064783461884007007523056946064899584\*a(n-60) -16294056286695850049547002274572240190  
 61) -333233305522203285352614541174833042241592032256\*a(n-62) +20597894616424076105661503966117871113  
 63) +206342303792645116186615933728367657989819990016\*a(n-64) -16444736449654546684080921843604854082  
 65) -90610830580669281648850701784053935924019621888\*a(n-66) +890174905189553585972248409318697190727  
 67) +27345115405290297188497153980462245068409798656\*a(n-68) -330845800076946812810543089271924616027  
 69) -5350062810598875580807935350063279580138209280\*a(n-70) +8330270004833621276238148417662208035177  
 71) +590807774385608600187792054379836607645089792\*a(n-72) -1371848721652188331182218108388444726854  
 73) -17554358576770937639357488522148218133938176\*a(n-74) +138550326488504911573703523315151801672531  
 75) -3626249233460961086387123169987888984096768\*a(n-76) -7621190173424905401435229631844950923542528\*  
 77) +421751512066698836117916913345490281562112\*a(n-78) +175498811870847501412698708269241288622080\*  
 79) -13876305971114000991959874100286837489664\*a(n-80) -502683252521624819998507914989785841664\*a(n-  
 81) +24181177809889217133507139098913013760\*a(n-82)

As of the “Last modified” line on the live entry (Oct 03 2025 15:40:13, revision 9) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 475a(n-1) - 97955a(n-2) + 11644948a(n-3) - 886260089a(n-4) + 44834024531a(n-5) - 14800169974a(n-6) + \dots \quad (1)$$

for all  $n > 81$ .

## 2 What the canonical-form clause counts

Fix the shape: 6 columns and  $L = n + 1$  rows, so  $L \cdot 6$  cells. Call a partition of the cells into nonempty classes a *pattern*; an array over an alphabet determines a pattern, two cells lying in the same class exactly when they carry the same value.

Every  $2 \times 2$  subblock of the array occupies two consecutive rows and two consecutive columns; writing those two rows as  $r$  and  $s$  the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix},$$

and the entry’s requirement on it is

$$\#\{\alpha, \beta, \gamma, \delta\} \in \{1, 2, 4\}. \quad (2)$$

The condition is stated purely in terms of equalities between entries, so it is invariant under any relabelling of the values and is therefore a property of the pattern alone. Call a pattern *admissible* when it has the property.

The clause “new values 0..3 introduced in row major order” selects, from each class of arrays that differ only by a relabelling, exactly one representative: reading the cells in row major order, the first occurrence of the value  $v$  must precede the first occurrence of  $v + 1$ . Every pattern with at most  $K = 4$  classes has exactly one such representative, and every array in canonical form is the representative of its own pattern. Hence, writing  $N_j(n)$  for the number of admissible patterns with exactly  $j$  classes,

$$a(n) = \sum_{j=0}^4 N_j(n). \quad (3)$$

**Lemma 1.** Let  $L_i(n)$  be the number of arrays over an alphabet of  $i$  letters satisfying the local condition, with no canonical-form clause. Then

$$L_i(n) = \sum_j N_j(n) i(i-1) \cdots (i-j+1), \quad 4! \sum_{j=0}^4 N_j(n) = \sum_{i=0}^4 \binom{4}{i} D_{4-i} L_i(n),$$

where  $D_m = m! \sum_{t=0}^m (-1)^t / t!$  is the number of derangements of  $m$  letters.

*Proof.* An array over  $i$  letters satisfying the condition determines an admissible pattern together with an injection of its classes into the alphabet, and conversely; a pattern with  $j$  classes admits  $i(i-1) \cdots (i-j+1)$  injections. That is the first identity. Writing  $i(i-1) \cdots (i-j+1) = j! \binom{i}{j}$  it reads  $L_i = \sum_j (j! N_j) \binom{i}{j}$ , so by binomial inversion  $j! N_j = \sum_i (-1)^{j-i} \binom{j}{i} L_i$ . Summing over  $j \leq 4$  and exchanging the order of summation,

$$\sum_{j=0}^4 N_j = \sum_{i=0}^4 L_i \sum_{j=i}^4 \frac{(-1)^{j-i}}{i! (j-i)!} = \sum_{i=0}^4 \frac{L_i}{i!} \sum_{t=0}^{4-i} \frac{(-1)^t}{t!},$$

and multiplying by  $4!$  turns the coefficient of  $L_i$  into  $\frac{4!}{i!} \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} (4-i)! \sum_{t \leq 4-i} (-1)^t / t! = \binom{4}{i} D_{4-i}$ , an integer.  $\square$

### 3 Each $L_i$ is a walk count, and the chain lumps

Fix  $i$  and let  $V_i = \{0, \dots, i-1\}^6$  be the possible rows. For  $r, s \in V_i$  declare  $r \rightarrow s$  an edge when placing  $s$  directly after  $r$  satisfies (2) at every column, and let  $M_i$  be the adjacency matrix. An array is exactly a sequence of rows, and its condition is exactly the conjunction of the edge conditions along that sequence, so

$$L_i(n) = \mathbf{s}_i^\top M_i^n \mathbf{1}.$$

Because the condition involves no row before the first, every row may start a walk, and  $\mathbf{s}_i = \mathbf{1}$ .

**Lemma 2.** Let  $\pi(r)$  be the equality pattern of the row  $r$ . For all  $r, r'$  with  $\pi(r) = \pi(r')$  and every pattern  $Q$ ,

$$\#\{s : r \rightarrow s, \pi(s) = Q\} = \#\{s : r' \rightarrow s, \pi(s) = Q\}.$$

Consequently the numbers  $\widehat{M}_i[P][Q]$  obtained by taking any representative of  $P$  are well defined, and with  $e_P = i(i-1) \cdots (i-|P|+1)$  the number of rows of pattern  $P$ ,

$$\mathbf{s}_i^\top M_i^m \mathbf{1} = \sum_P \varepsilon_P f_m(P), \quad f_0 \equiv 1, \quad f_{m+1}(P) = \sum_Q \widehat{M}_i[P][Q] f_m(Q),$$

where  $\varepsilon_P = e_P$  if rows of pattern  $P$  may start a walk and 0 otherwise.

*Proof.* If  $\pi(r) = \pi(r')$  then the map sending the value of  $r$  at a column to the value of  $r'$  at that column is a well-defined injection between the letters used, and extends to a permutation  $\sigma$  of the alphabet with  $r' = \sigma(r)$ . The edge condition is a statement about equalities among entries, so  $r \rightarrow s$  if and only if  $\sigma(r) \rightarrow \sigma(s)$ ; and  $\pi(\sigma(s)) = \pi(s)$ . Hence  $s \mapsto \sigma(s)$  is a bijection between the two successor sets preserving patterns, which is the first claim. It says the chain is strongly lumpable over  $\pi$ , so  $(M_i^m \mathbf{1})(r)$  depends on  $r$  only through  $\pi(r)$ ; calling that value  $f_m(\pi(r))$  gives the recursion, and summing over  $r$  in each class gives the displayed identity, the number of rows of pattern  $P$  being the number of injections of its  $|P|$  classes into the alphabet.  $\square$

Assembling the 4 lumped chains into one block-diagonal matrix  $N$  of size  $S = 342$  and putting the weights of Lemma 1 into the start vector,  $\mathbf{v} = (\binom{4}{i} D_{4-i} \varepsilon_P)_{i,P}$ , gives

$$4! a(n) = \mathbf{v}^\top N^n \mathbf{1}. \quad (4)$$

The unlumped alphabet had 4890 rows in all; the lumped chain has  $S = 342$  states.

## 4 The criterion, and the computation

Let  $q(t) = t^{82} - \sum_i c_i t^{82-i}$  be the characteristic polynomial of (1).

**Lemma 3.** *For every  $n$  with  $m = n + 1 - 1 \geq 82$ ,*

$$4! \left( a(n) - \sum_i c_i a(n-i) \right) = \mathbf{v}^\top N^{m-82} q(N) \mathbf{1}.$$

Hence (1) holds from that point on if and only if  $\mathbf{v}^\top N^j q(N) \mathbf{1} = 0$  for every  $j \geq 0$ , and it is enough to check  $j = 0, 1, \dots, S-1$ .

*Proof.* Substitute (4) into the left side and factor  $N^{m-82}$  out of  $N^m - \sum_i c_i N^{m-i}$ . The scalar  $4!$  is nonzero, so it does not affect whether the left side vanishes. For the last clause put  $w = q(N) \mathbf{1}$ : by Cayley–Hamilton  $N^S$  is an integer combination of  $I, N, \dots, N^{S-1}$ , so each  $\mathbf{v}^\top N^j w$  with  $j \geq S$  is the corresponding combination of earlier values.  $\square$

**Theorem 4.** *The recurrence (1) holds for every  $n > 81$ , which is the statement of Section 1.*

*Proof.* The lumped transition counts were obtained by enumerating, for one representative row of each pattern, all admissible successors and sorting them by pattern — the successors being generated column by column, since the edge condition is a conjunction of one constraint per column. The vector  $w = q(N) \mathbf{1}$  was then formed by 82 matrix–vector products in exact integer arithmetic and  $\mathbf{v}^\top N^j w$  evaluated for  $j = 0, 1, \dots$ , again exactly. Every one of those integers vanishes from the index corresponding to  $n = 81 + 1$  onwards, and the run continued until  $w$  was the zero vector, which settles all larger  $j$  at once. Lemma 3 gives the theorem.  $\square$

## 5 Verification

Three checks, each independent of the algebra above.

First, the whole construction was compared against the entry’s DATA: the right-hand side of (4), divided by  $4!$ , was evaluated at every one of the 11 published indices and agrees exactly. This is a strong test of the modelling and not only of the arithmetic — the inversion of Lemma 1, the reading of the local condition, and the alignment of the entry’s index with the number of rows would each show up as a mismatch, and the quotient by  $4!$  came out an integer at every index.

Second, the lumped chain was checked against the unlumped one: for the small shapes the walk counts  $\mathbf{s}_i^\top M_i^m \mathbf{1}$  were computed directly over the full alphabet as well and agree with  $\sum_P \varepsilon_P f_m(P)$  term by term.

Third, the recurrence (1) was evaluated on the published terms in exact integer arithmetic with no matrices involved, and the annihilation test of Lemma 3 was run until the working vector was identically zero rather than to a sampled prefix.

## References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A210058.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7; Stirling and falling-factorial inversion, Section 1.9.)
- [3] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, Springer, 1976. (Strong lumpability, Section 6.3.)
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.